

Who will use the coupon?

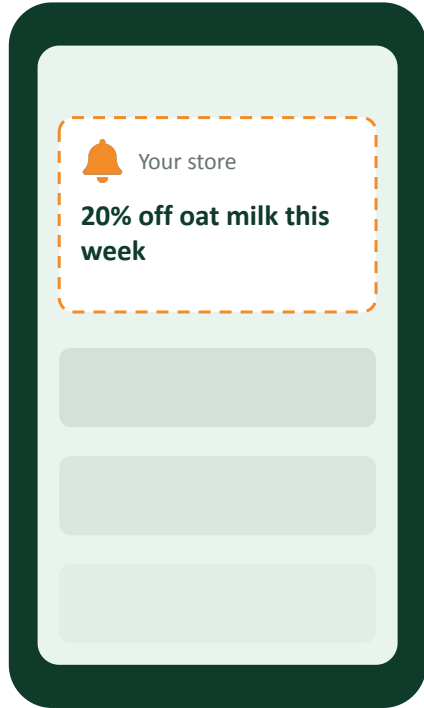
Agentic AI Marketing Decision Lab: predictive modeling and AI decision support for retail campaigns

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You get a coupon. Someone chose to send it.



What you see

Nice discount!

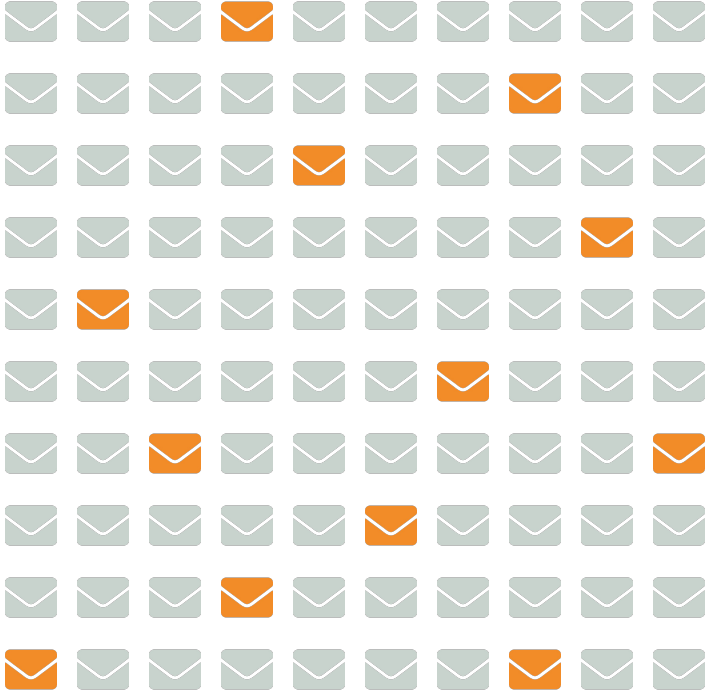


What the store sees

Money given away. Did it change what you bought, or would you have bought it anyway?

Send it to the wrong people, and the budget is wasted.

Most coupons go nowhere



1 in 8

coupon mailings in a real grocery dataset were used even once

And some of those shoppers would have bought anyway.

Source: dunnhumby "The Complete Journey": 889 of 7,208 mailings led to a redemption.

Two shoppers, one coupon



20% off oat milk

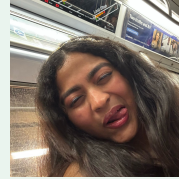


PRNAV

Buys oat milk every week

Uses coupon, but would have bought anyway.

The coupon changed nothing.



SHREYA

Has never tried oat milk

Uses coupon, likes it, and keeps buying.

The coupon made the difference!

Same coupon, different value. Both show up as "used." Telling them apart is our job.

Marketers sort customers into four groups

Buys WITH the coupon?

Yes

Persuadables

Buy only with coupon

SHREYA



Sure things

Buy anyway

PRANAV



No

Lost causes

Won't buy either way

CLAUDIA

Sleeping dogs

The offer puts them off

ANNA

No

Yes

Buys WITHOUT the coupon?



The coupon should go to Shreya.

But a model that only asks "who will buy?" ranks Pranav first.

Framework: Radcliffe & Surry (2011)

Three questions we want to answer



QUESTION 1

Will they use it?

Will this household use a coupon from this campaign?



QUESTION 2

What is it worth?

If they use it, how much does that bring the store?



QUESTION 3

Did it make a difference?

Would they have bought it anyway, without the coupon?

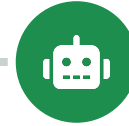
Together, they tell a marketer who is worth a coupon.

How the field got here

FOR DECADES

SINCE AROUND 2000

NOW



Who will respond?

Direct marketers predict response from past behavior: how recently, how often, how much someone buys.

Who will change?

Uplift modeling predicts the difference a coupon makes, not just who buys.

Can AI help run it?

AI agents can automate parts of the analysis and explain results in plain language.

Our project brings all three together in one tool.

e.g. Bult & Wansbeek (1995); Fader, Hardie & Lee (2005) | Radcliffe & Surry (2011); Künzle et al. (2019) | Guo et al. (2024), DS-Agent; Hong et al. (2025), Data Interpreter

The data: real grocery shopping records



2,500

households that shop
there often



2 years

of every purchase they
made



2.6M

items on their receipts



7,208

coupon mailings sent

It records what people bought, which coupons they received, and which they used.

Households appear only as ID numbers. Basic household details are known for about 1 in 3.

Source: dunnhumby, "The Complete Journey" (public dataset)

What we are building for TD Bank



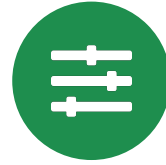
1 Prepare

AI agents clean and organize two years of records



2 Predict

Models answer our three questions for every household



3 Decide

A dashboard where a marketer picks a campaign and sees who to target



4 Explain

An AI assistant answers "why?" in plain English

Goal: send the right coupon to the right people, and know it worked.

Why this is harder than it sounds



We never see the other version

Shreya used the coupon. We can't rewind and see what she'd have done without it. Every answer to "did it make a difference" is an estimate.



Coupons weren't sent at random

Stores tend to mail their best customers, so those who got coupons already shop more.



AI answers must be trustworthy

Every number the assistant gives must match what the models actually say.

These are the problems we will work through this semester.

Team 6



**Not just who will use the coupon,
but who it actually changes.**

Thank you. Questions?

Yuxin Cai · Shreya Shetty · Pranav Jain · John Won · Zhiyan Yang

Sponsored by TD Bank | Data: dunnhumby, The Complete Journey